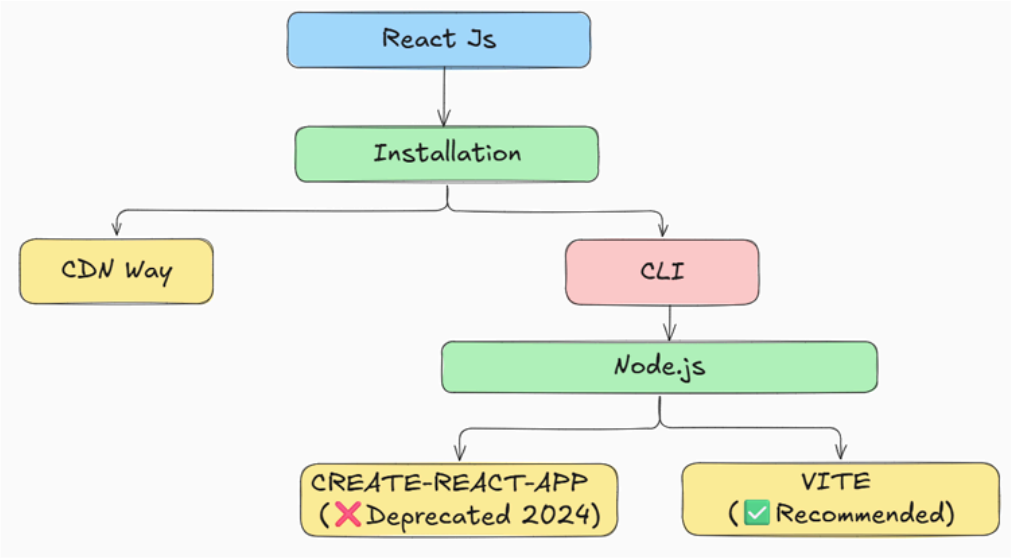
4. Installation of React JS

▼ There are multiple ways we can install the react application:



▼ What is Vite?

- Vite is a Build Tool.
- Vite (French word for “quick”) is a build tool that aims to provide a faster and leaner development experience for modern web projects.

▼ Why use Vite instead of React?

- Vite offers some advantages over Create-React-App, among them: Faster Development Server.
- Vite uses native ES Modules to create a development server that is quicker and more effective.

▼ Installation Process by Using Vite+React:

- Vite is frontend build tool and development server.
- Vite is a faster and lightweight.
- It provides better performance and a smoother developer experience.

▼ Step-1: Install Node.js

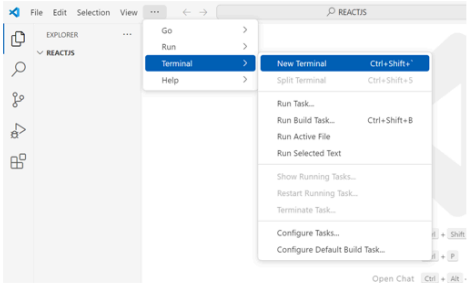
- Before installing React, make sure Node.js is installed on your system.
- After installation open command prompt (cmd) to check the version of installed node .js
- Command to check the node.js version: `node -v`
- Command to check the npm version: `npm -v`
- 👉 If not installed, download it from: https://nodejs.org/

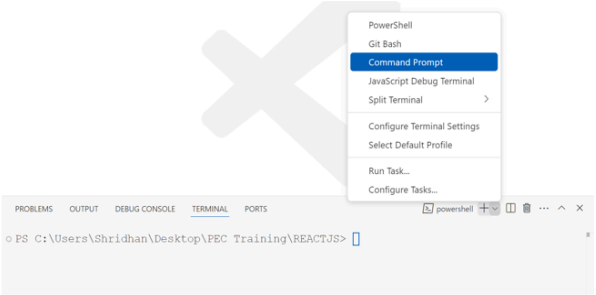
▼ Step-2: Create a New Folder

- Create a new folder and give a name to folder

▼ Step-3: Open this folder inside the vs code

- Click Terminal → New Terminal
- Click (+) icon and open the command prompt not Powershell.





▼ Step-4: Run the following command in cmd

- Command:

💡 npm create vite@latest project-name

- And just hit enter on keyboard.

▼ Step-5: Select a framework

- After giving name it will ask the select a framework.
- Only Select → **React**
- How to select ? → Use down arrow key on keyboard for selecting option react option.
- **For Example** → Select a framework: `React`

```
◆ Select a framework:
  ○ Vanilla
  ○ Vue
  ● React
  ○ Preact
  ○ Lit
  ○ Svelte
  ○ Solid
  ○ Ember
  ○ Qwik
  ○ Angular
  ○ Marko
  ○ Others
```

▼ Step-6: Select a variant

- After selecting a framework it will ask for you variant.
- Variant is nothing but which language you are going to use in react js.

- How to select ? → Use down arrow key on keyboard for selecting option javascript option.
- **For Example** → Select a variant: Javascript (✅)
- Don't select Javascript+React Compiler (❌)

```
◆ Select a variant:
  ○ TypeScript
  ○ TypeScript + React Compiler
  ● JavaScript
  ○ JavaScript + React Compiler
  ○ RSC
  ○ React Router v7 ↗ https://reactrouter.com
  ○ TanStack Router ↗ https://tanstack.com/router
  ○ RedwoodSDK ↗ https://rwsdk.com
  ○ Vike ↗ https://vike.dev
```

▼ Step-7: Install with npm and start now?

- When you run `npm create vite@latest`, Vite only **creates the folder** and project files. (like Auto install + Auto run)
- This step asks:

Should Vite automatically:

1. Run `cd project-folder`
2. Run `npm install` (install all dependencies)
3. Run `npm run dev` (start the app)

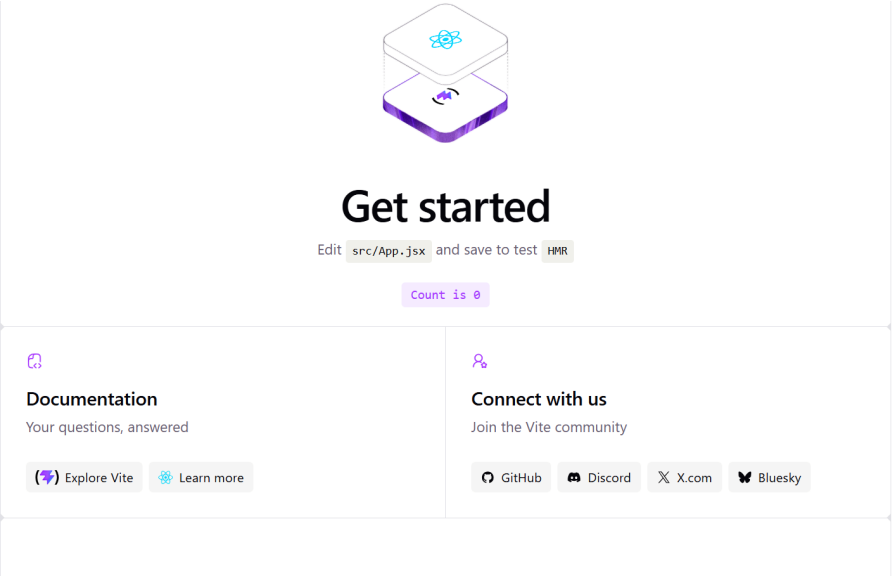
If you choose:

- `Yes` → Vite will automatically install everything and start the development server.
- `No` → You will need to manually run:
- **So what should you choose? →Select →** `Yes`
- It saves time and makes setup easier.

```
◆ Install with npm and start now?
  ● Yes / ○ No
```

▼ Step-9: Command to run the React JS application

- To run the react application, we have to use the following command:

💡 npm run dev

- It will run on the port on → http://localhost:5173/